Continuous Planar Isotropy and the Pythagorean Formula

A Plain-English Explanation of Lean-Formalized Hypotheses and Proofs

AI-generated mathematical exposition: **ChatGPT (GPT-6, High thinking effort)**
Research question and initiating prompt: **Alessandro Odasso**

Abstract

Why should the length of a displacement with coordinates (x, y) be $\sqrt{x^2 + y^2}$? In this manuscript, length begins as an arbitrary function, not as that familiar formula. We explain two related implications formalized in Lean 4: a conditional Pythagorean theorem using rigid motions, reflection, and axis reachability; and a stronger characterization in which continuity of a linear planar action *derives* axis reachability and reflection symmetry. An explicit trigonometric model proves the converse, while the maximum norm shows why finite rotational symmetries alone are insufficient. The proofs are conditional statements about explicitly assumed linear motions; they do not construct geometry from incidence axioms or establish historical novelty.

1 Research prompt, authorship, and scope

AI and prompt disclosure. Alessandro Odasso initiated the research question by asking OpenAI's ChatGPT whether it could find an *original proof of the Pythagorean theorem*. The manuscript text and proposed mathematical argument were generated with ChatGPT (GPT-6, High thinking effort); the accompanying formalization and subsequent extensions were developed with AI assistance and checked in Lean 4/Mathlib. The word *original* reports Odasso's objective, not a verified claim of priority, novelty, or originality in the mathematical literature. Lean verifies formal implications from stated hypotheses, not such historical claims.

This text is an accessible explanation of what the repository actually formalizes. Its logical starting point is not a triangle diagram or the distance formula: it is an unknown function of two coordinates and an explicitly described family of linear transformations. The key mathematical point is to keep *length* separate from the coordinate expression that the proof eventually forces.

2 The objects and assumptions

Represent a displacement by $(x, y) \in \mathbb{R}^2$. Let

$$L : \mathbb{R}^2 \longrightarrow \mathbb{R}$$

be a *primitive length function*. Initially there is no definition $L(x, y) = \sqrt{x^2 + y^2}$, no norm axiom, and no assumed continuity of L.

Each planar map below is linear and is represented by a real 2×2 matrix. Write F_t for a one-parameter family of such maps and Q for a distinguished quarter-turn. The assumptions of the main characterization are:

1. **Action law.** For all s, t, $F_{s+t} = F_s \circ F_t$.
2. **Primitive oriented quarter-turn.** $Q(1, 0) = (0, 1)$ and $Q^2(x, y) = (-x, -y)$ for all x, y. The matrix of Q is *not* assumed in advance.
3. **Membership.** There is a real parameter θ for which $F_\theta = Q$.

4. **Continuous action coefficients.** The first-column entries $t \mapsto (F_t)_{11}$ and $t \mapsto (F_t)_{21}$ are continuous.

5. **Axial calibration.** $L(z, 0) = |z|$ for every real z.

6. **Rigidity.** $L(F_t(x, y)) = L(x, y)$ for every t, x, y.

These premises assume a sufficiently rich *linear* motion action; they do not infer its existence from incidence or congruence alone. In contrast, we do *not* assume that L is positive, continuous, homogeneous, subadditive, or symmetric under reflection. The latter properties will follow from identifying L.

Theorem 2.1 (Characterization formalized in Lean)**.** *For every function L satisfying axial calibration,*

$$\begin{aligned} &\textit{there exist } F, Q \textit{ satisfying the action, quarter-turn,} \\ &\textit{membership, continuity, and rigidity hypotheses} \\ \iff \qquad & L(x, y) = \sqrt{x^2 + y^2} \quad \textit{for all } x, y. \end{aligned}$$

In particular, under those hypotheses every displacement with $c = L(a, b)$ satisfies $a^2 + b^2 = c^2$.

The equivalence is the Lean theorem `continuous_isotropy_iff_euclidean_length`. We explain its forward and reverse implications in the order used by the proof modules.

3 Forward implication: five proof steps

Step 1. The quarter-turn and the shape of every motion

Applying the square law to $(1, 0)$ gives

$$Q(0, 1) = Q^2(1, 0) = (-1, 0).$$

Linearity then forces

$$Q(x, y) = (-y, x), \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

This matrix is a *conclusion*, not a premise. By the action law and $F_\theta = Q$,

$$F_t Q = F_{t+\theta} = F_{\theta+t} = Q F_t.$$

A real 2×2 matrix commuting with J must have the form

$$F_t = \begin{pmatrix} u(t) & -v(t) \\ v(t) & u(t) \end{pmatrix}, \qquad F_t(x, y) = (u(t)x - v(t)y,\ v(t)x + u(t)y). \tag{1}$$

The action law and the invertibility of Q also give $F_0 = I$, so $u(0) = 1$, $v(0) = 0$. These deductions correspond to `primitive_quarter_turn_is_J`, `primitive_quarter_turn_forces_flow_shape`, and `action_zero_of_primitive_quarter_turn`.

Step 2. Continuity excludes radial scaling

Define the nonnegative quantity

$$s(t) = u(t)^2 + v(t)^2.$$

Multiplying matrices of the form (1) yields

$$s(t + r) = s(t)s(r), \quad s(0) = 1, \quad s(\theta) = 1. \tag{2}$$

Consequently $s(t+\theta) = s(t)$. The parameter θ cannot be zero because $F_0 = I$ whereas $Q(1,0) = (0,1)$. By the coefficient continuity, s is continuous with a nonzero period, hence has a compact range and a greatest value $M \geq 1$.

Choose r with $s(r) = M$. For every t,

$$s(t)M = s(t+r) \leq M,$$

so $s(t) \leq 1$. But (2) also gives $s(t)s(-t) = s(0) = 1$, and applying the same bound to $-t$ implies $s(t) \geq 1$. Therefore

$$\boxed{u(t)^2 + v(t)^2 = 1 \quad \text{for every } t.} \tag{3}$$

Expanding the transformed coordinates now shows

$$(ux - vy)^2 + (vx + uy)^2 = (u^2 + v^2)(x^2 + y^2) = x^2 + y^2. \tag{4}$$

This does *not* use L, its reflection symmetry, or any derivative. In Lean the main ingredients are `periodic_nonnegative_character_eq_one` and `continuous_action_preserves_quadratic`.

Step 3. Every displacement reaches the positive axis

For a fixed displacement (a, b), consider the second coordinate of $F_t(a, b)$. By (1) it is the continuous function

$$f(t) = v(t)a + u(t)b.$$

At $t = 0$ its value is b. At $t = 2\theta$, the action gives $F_{2\theta} = Q^2 = -I$, so its value is $-b$. The intermediate value theorem therefore provides a parameter T with $f(T) = 0$. Write $F_T(a, b) = (z, 0)$. If $z < 0$, composing with the half-turn $F_{2\theta}$ changes $(z, 0)$ to $(-z, 0)$. Thus in every case

$$\exists T, z \quad z \geq 0, \qquad F_T(a, b) = (z, 0). \tag{5}$$

The formal result is `axis_reachable_of_continuous_action`. This is a derived fact, not an extra axiom in the continuous theorem.

Step 4. Compare the two descriptions of the axis point

Equation (4) applied to (5) gives

$$z^2 = a^2 + b^2.$$

On the other hand, rigidity and axial calibration say

$$L(a, b) = L(F_T(a, b)) = L(z, 0) = |z| = z.$$

Because $z \geq 0$,

$$\boxed{L(a, b) = \sqrt{a^2 + b^2}}, \qquad \boxed{a^2 + b^2 = L(a, b)^2}. \tag{6}$$

The length formula has been *proved* under the action assumptions, rather than used to define L. This is `length_eq_sqrt_without_mirror`. The identity in (6) is the coordinate Pythagorean formula for the displacement whose primitive length is $c = L(a, b)$.

Step 5. Mirror symmetry is a consequence

Formula (6) immediately yields

$$L(x, -y) = \sqrt{x^2 + (-y)^2} = \sqrt{x^2 + y^2} = L(x, y).$$

Thus mirror symmetry, unlike in the earlier conditional proof, is not a hypothesis. Lean records this as `mirror_symmetry_of_continuous_isotropy`.

4 Reverse implication: an explicit model

The reverse direction does not merely assert that such motions ought to exist. For $L(x, y) = \sqrt{x^2 + y^2}$, set

$$F_t = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}, \qquad Q = F_{\pi/2}.$$

The trigonometric addition identities prove the action law; sine and cosine are continuous; Q has the required quarter-turn behavior; and the usual identity $\sin^2 t + \cos^2 t = 1$ proves rigidity. The Lean definitions `standardRotation`, `standardQuarterTurn`, and the theorem `continuous_isotropy_model_of_euclidean_length` formalize this converse. This construction uses ordinary real analysis; it is not a derivation of trigonometry from synthetic geometry.

5 How the earlier conditional theorem fits

The archived Lean theorem `pythagoras_from_metric_free_quarter_turn` does *not* require continuity of the action. Instead, in addition to the action, quarter-turn, membership, axial calibration, and rigidity, it explicitly assumes:

- *Mirror symmetry:* $L(x, -y) = L(x, y)$;
- *Axis reachability for the selected vector:* $F_T(a, b) = (z, 0)$ for some $z \geq 0$;
- *Length identification:* $c = L(a, b)$.

Its proof obtains the normal form (1) from the quarter-turn. Instead of Step 2's compactness argument, mirror symmetry and rigidity show

$$L(u, v) = 1, \quad L(u, -v) = 1, \quad L(u^2 + v^2, 0) = 1.$$

Calibration and $u^2 + v^2 \geq 0$ imply $u^2 + v^2 = 1$. With the *assumed* axis reachability, the algebraic calculation in Step 4 yields $a^2 + b^2 = c^2$.

The two theorems therefore trade assumptions: the older theorem permits a discontinuous action but requires mirror symmetry and an axis hit; the newer theorem requires continuity of two action coefficients and *derives* both consequences. This is not a claim that every single assumption in one statement is weaker than the corresponding assumption in the other.

6 Why one quarter-turn is not enough

Consider the maximum norm

$$B(x, y) = \max\{|x|, |y|\}.$$

It is continuous, axially calibrated, positive definite, homogeneous, and satisfies the triangle inequality. It is also invariant under reflection and the four coordinate quarter-turns. But

$$B(1, 1) = 1, \qquad 1^2 + 1^2 = 2 \neq B(1, 1)^2.$$

So a finite collection of elementary symmetries cannot by itself force the Euclidean length formula.

Lean proves more: among matrices of the form (1), the only B-rigid ones are the four signed coordinate quarter-turns. A real additive action containing Q would also contain a rigid square root $F_{\theta/2}$ of Q; no such matrix exists for B. Hence no B-rigid additive real linear action containing the primitive quarter-turn exists, *even without* a continuity hypothesis. The formal statements include `boxLength_rigid_normal_iff`, `boxLength_not_pythagorean`, and `boxLength_no_additive_quarter_turn_action`. This example is an obstruction to *finite* symmetries, not a counterexample satisfying all the hypotheses of either main theorem.

7 Infinitesimal interpretation and proof boundaries

If the motion coefficients have the stated derivatives at zero, the infinitesimal generator has the form ωJ after the radial derivative is shown to vanish. Algebraically, for a symmetric matrix M, the infinitesimal invariance equation $J^T M + MJ = 0$ forces $M = \lambda I$. The repository contains separate Lean statements on the generator, conserved quadratic expressions, and the classification of infinitesimally invariant symmetric forms. These *supplement* the group argument above; differentiability and Lie theory are not needed for Theorem 2.1.

The precise boundary matters. Lean establishes that the displayed conclusions follow from formal hypotheses, and the explicit model shows joint satisfiability in standard real analysis. It does not prove that the action exists from bare incidence or primitive congruence axioms, establish a minimal axiom system, or certify historical novelty. The principle that rich linear isometry groups select Euclidean structure has substantial classical context. A kernel-level axiom audit concerns Lean's logical dependencies, not whether these *geometric* hypotheses are minimal.

8 Where to find the formal proofs

The proof steps above correspond to the following files in the `palomar-preparation` branch:

- `PythagoreanIsotropy.lean`: primitive quarter-turn, motion normal form, and the archived conditional theorem;
- `ContinuousIsotropy.lean`: derived axis reachability;
- `CompactCharacter.lean`: the periodic-character argument;
- `CompactIsotropy.lean`: quadratic preservation and length identification without mirror symmetry;
- `EuclideanModel.lean` and `IsotropyCharacterization.lean`: explicit model and the characterization equivalence;
- `IsotropyCounterexamples.lean`: maximum-norm obstruction.

Repository: pythagorean-isotropy-lean (palomar-preparation branch). The root `README.md` provides reproducible toolchain, Comparator, verification, archival and possible Palomar publication details; those operational details are deliberately kept out of this mathematical exposition.